National University of Computer and Emerging Sciences

Lahore Campus

Data Communication & Networks(EE2007)

Sessional-I Exam

Total Time (Hrs.): 1

Total Marks: 20

Total Questions: 4

Date: September 24, 2025

Course Instructor(s)

Mr. Khalid Ijaz

Roll No _____ Section _____ Student Signature _____

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Attempt all the questions.

CLO 1: Describe Internet architecture and network performance parameters.

Q1: **Describe** how the OSI model operates when a user streams a live online lecture. **Describe** the role of each layer in ensuring smooth and reliable delivery of audio and video data. Also, **describe** the protocols in brief that can be used by the application layer to provide above service. [8 marks]

CLO 2: Explain physical layer concepts in data communication including data rates, line coding schemes, and Shannon's capacity limits.

Q2: An image is 1600 × 1200 pixels with 3 bytes/pixel. Assume the image is uncompressed. **Explain** how much transmission time is required to transmit an image over a 56-kbps DSL modem channel. [3 marks]

CLO 2: Explain physical layer concepts in data communication including data rates, line coding schemes, and Shannon's capacity limits.

Q3: The signal-to-noise ratio is often given in decibels. Assume that SNR_{db} = 36 and the channel bandwidth is 2 MHz. Explain what the theoretical channel capacity is for the noisy channel. [3 marks]

CLO 3: Analyze the design principles involved in data link layer.

Q4: **Analyze** the generation of the Hamming code for the 4-bit data 1001 using even parity. Assume the receiver receives the code 0011101. Analyze the complete process of the Hamming technique to detect error. [6 marks]

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